Question 1

Explain the difference between EPC (4G) and 5GC (5G).

- **EPC (Evolved Packet Core)** is the core network architecture used in **4G LTE** networks. It was designed primarily for mobile broadband and high-speed data services.
- **5GC (5G Core)** is the cloud-native core network used in **5G**. It supports ultra-low latency, massive IoT connectivity, automation, and AI-driven services. It is more flexible and uses a service-based architecture (SBA) that allows network functions to communicate through APIs.
- The presentation notes that **4G focuses on high-speed broadband**, while **5G supports ultra-low latency, IoT, and automation**, reflecting the capabilities enabled by the 5G Core.

Question 2

Why is network slicing important in 5G? Provide real-world examples.

Network slicing allows a single 5G network to be divided into multiple virtual networks, each optimized for a specific use case.

Importance:

- Improves efficiency by allocating resources based on application needs.
- Allows different services to operate with different performance requirements on the same physical network.

Examples:

1. **Autonomous vehicles** – require ultra-low latency and high reliability.
2. **Smart factories** – need reliable machine-to-machine communication.
3. **IoT sensors** – require support for massive numbers of low-power devices.
4. **Video streaming** – needs high bandwidth for HD and 4K content.

These examples align with the presentation's discussion of **IoT, automation, drones, and autonomous vehicles as key 5G applications.**

Question 3

What is Multi-access Edge Computing (MEC), and how does it reduce latency?

MEC places computing resources closer to users at the network edge instead of sending all data to distant cloud servers.

How it reduces latency:

- Data travels a much shorter distance.
- Processing occurs near the device rather than in a centralized cloud.
- Results are returned faster, which is critical for real-time applications.

Examples:

- Self-driving cars
- Industrial robots
- Augmented/Virtual Reality
- AI-powered video analytics

This supports the presentation's emphasis on **5G ultra-low latency and AI/IoT applications.**

Question 4

Describe the role of AI in dynamic resource allocation.

AI continuously monitors network conditions and automatically adjusts resource allocation to improve performance.

AI can:

- Predict traffic congestion.
- Allocate bandwidth where demand is highest.
- Optimize wireless spectrum usage.
- Improve quality of service for critical applications.
- Reduce network operating costs.

The presentation specifically mentions **AI-based monitoring and optimization of networks**, showing how AI helps manage network resources dynamically.

Question 5

In SBA, what is the role of the Network Repository Function (NRF)?

In the **Service-Based Architecture (SBA)** of the 5G Core, the **Network Repository Function (NRF)** acts as a service registry.

Functions of the NRF:

- Keeps track of available network functions.
- Allows network functions to discover and communicate with one another.
- Maintains information about network services and their status.
- Helps enable flexible, cloud-native operation of the 5G Core.

Think of the NRF as a **directory or phone book** that helps 5G network functions find each other and exchange services efficiently.

Source: The uploaded presentation provides the background on 5G core networking and AI-enabled network architecture, though NRF itself is not explicitly defined in the slides.